A DIFFERENTIAL GRADED MODULE WITH PROPERTY (P) IS K-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: dependencies carried as navigation stubs: the property (P) short exact sequence (Lemma lemma-property-P-sequence) and the free-shift vanishing (Lemma lemma-hom-from-shift-free), both in-source; the triangle reduction cites the standard Derived Categories homological lemma as external tooling.

Editorial note. The label lemma-property-P-K-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Property (P) filtration. Same unit, the definition of property (P) and Lemma lemma-property-P-sequence. The proof takes the filtration F_bullet on P given by property (P) and uses the short exact sequence of Lemma lemma-property-P-sequence to produce a distinguished triangle at each filtration step.
- Free case. Same unit, Lemma lemma-hom-from-shift-free. Final step (source line 2487): Hom in K of a shift A[k] into an acyclic module vanishes, which follows from Lemma lemma-hom-from-shift-free.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Prerequisite pointer invoked at proof line(s) 2473; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K1: sources/stacks-dga/pinned-original-source.tex, source lines 2118–2118. Prerequisite pointer invoked at proof line(s) 2486; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KL: sources/stacks-dga/pinned-original-source.tex, source lines 2425–2425. Prerequisite pointer invoked at proof line(s) 2471; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014S: sources/stacks-dga/pinned-original-source.tex, source lines 1347–1347. Prerequisite pointer invoked at proof line(s) 2469; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module with property (P). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(P, N) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 014S). Let $F_\bullet$ be a filtration on P as in property (P). The short exact sequence of Lemma 09KL produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(F_iP, N) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules F_iP has a finite filtration by admissible monomorphisms, whose graded pieces are direct sums of shifts $A[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(A[k], N) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 09K1. □